

Enterprise Computing Data Visualisation Task

Year 12 Higher School Certificate Enterprise Computing Task 2



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AGRICULTURAL HIGH SCHOOL

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Acknowledgement

Dataset: <https://www.kaggle.com/datasets/gabrielramos87/an-online-shop-business>

This dataset, "E-commerce Business Transaction" was retrieved from Kaggle, and is a year's worth of sales transactions for a UK based e-commerce store. It is an unnamed shop based in London, and it has both domestic and international customers. Its customers range from individuals to small businesses buying in bulk for resale. Overall it contains around 500,000 records in total. The dataset was uploaded by Gabriel Ramos, under a copyright free license. Based on the description provided on the website it seems that this dataset is aimed for an educational use. It was pulled directly from the business's database system. The business remains unnamed, and sensitive customer information such as names, addresses and genders were removed to maintain confidentiality.

In regards to the data's reliability, the data can be seen as reliable. This dataset is accurate, free from errors and consistent throughout, but is not in 3NF (Third Normal Form). Given that this is a legitimate business, the creator was unable to provide a public source due to the need to maintain the privacy of the business and its customers. This in turn makes the dataset trustworthy, as the deliberate removal of sensitive data demonstrates that proper data governance was enacted. The data is complete, with all critical information present. Therefore, the data is reliable.

For validity, the data appears to be valid. This dataset represents an e-commerce store's transactions over a year's period of time. It uses unique identifiers to define each transaction, with "C" representing a cancellation. It is formatted correctly, with fields such as date being validated. Therefore this dataset is valid as it accurately represents the business.

There would be little collection bias in this dataset as it was pulled directly from the business database system. However, one potential source of bias may be in its data representation of a single business. As it is unnamed, findings may not generalise to the broader e-commerce industry.

Research and Planning

Reliability and Relevance

This dataset comes from Kaggle, a Google owned online platform that serves as a repository for data science and machine learning. Its creator, Gabriel Ramos, is the Data Analytics Consultant/Freelancer based in London, England. He has uploaded three different datasets, all three being labelled as copyright free public data. The business this dataset represents has been anonymised, with sensitive customer information removed before publication. The data is highly credible with over 26,000 downloads.

The dataset contains over 500,000 unique records, all with the following information: TransactionNo, Date, ProductNo, ProductName, Price, Quantity, CustomerNo and Country. There are no missing fields, making the data consistent. This removes analytical error and the data has standardised formatting as well as a concise data dictionary for documentation.

This dataset cannot be directly verified against its original source, as the business remains unnamed with no public source provided. However, given that the data was extracted directly from the business's own database system, there is little chance for manual entry errors. The dataset formatting is consistent and aligns with what a typical business transaction database would look like. General verification can be conducted by comparing the data against other UK e-commerce sales data to see if they have correlation. Additionally, product pricing can be verified against other publicly available prices for similar goods.

In order to find supporting data for this dataset, it can be compared to online, named datasets. For example, the Office of National Statistics (ONS) is a UK government agency that provides public data. This can be used to either support or oppose this dataset. This dataset can also be analysed alongside other e-commerce datasets from Kaggle during the same time period (Dec. 2018 to Dec. 2019).

Limitations and Uncertainties

This dataset only covers the equivalent of one year (Dec. 2018 to Dec. 2019). This makes it impossible to identify multi year trends, seasonal shifts and long term shifts in customer behaviour. Conclusions that attempt to make predictions about future years should be treated with caution, as this single year cannot be entirely representative of the broader business performance. To solve this, other datasets could be used from Kaggle or ONS to spot correlations, and then make informed predictions based on that information.

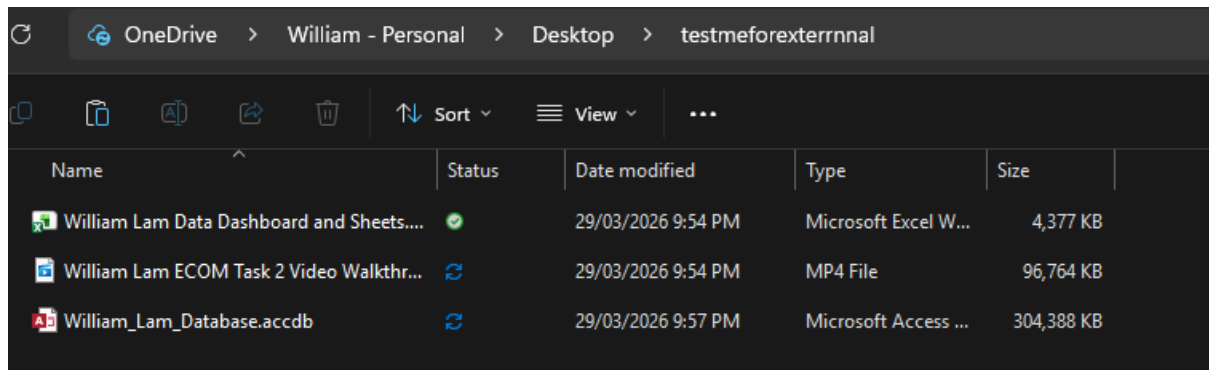
This dataset represents a B2C (Business to Consumer) e-commerce store, and such will have a varying demographic of customers. However the dataset has been anonymised, and as such trends relating to customer demographics cannot be drawn. This customer bias can lead to skewed metrics between normal consumers and bulk buyers. However this can be solved by getting the sensitive customer information for analysis. Sample bias cannot be properly shown and hence dealt with, which can lead to misinformed results and predictions.

There is a low risk of collection bias, as the data was extracted using digital systems directly from the business database.

This dataset covers December 2018 to December 2019. A single year is not entirely representative of the business's typical performance. Additionally, this predates the COVID-19 pandemic, which led to a massive increase in online shopping. Additionally, after COVID, e-commerce remained well above pre-COVID, so this would have substantially impacted the business. As such, this dataset cannot represent the business accurately for future predictions. A solution to this would be to limit the scope of the investigation to just the 2018-2019 period, hence making this dataset highly relevant. It could act as a pre-COVID set when compared with post COVID to view insights and how the industry has changed as a whole.

Import Instructions

When downloading both the database and excel spreadsheet, please ensure the files are located in the same directory/folder. This is to maintain the live connection between them.

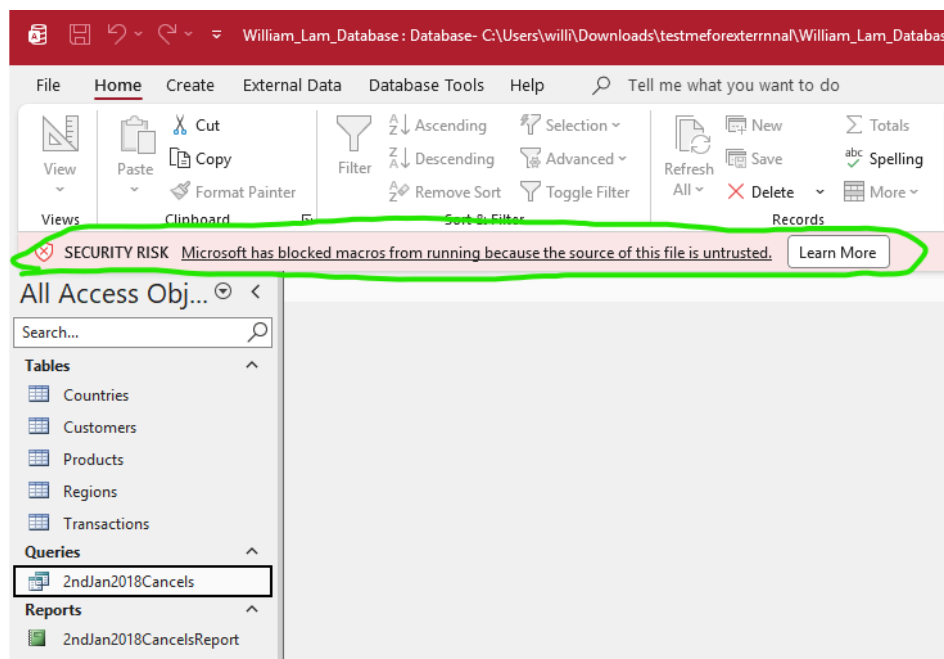


Both Excel and Access files in the same folder

When opening the files, please ensure that all external content is enabled. If there are errors regarding them, just click ok and then allow.

Access

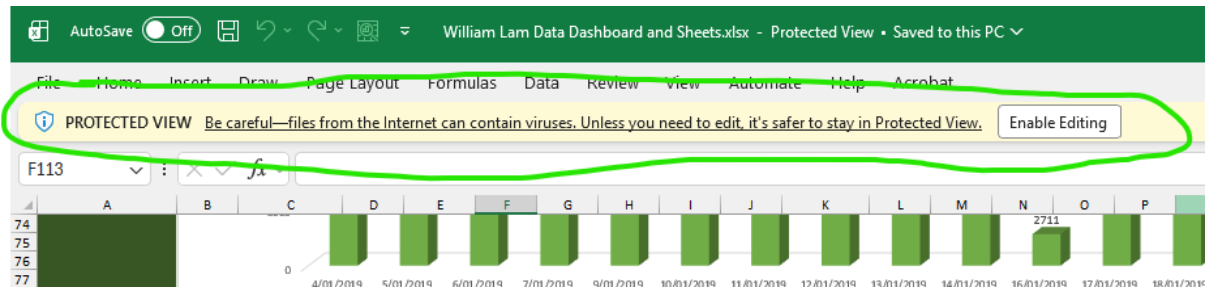
If this occurs:



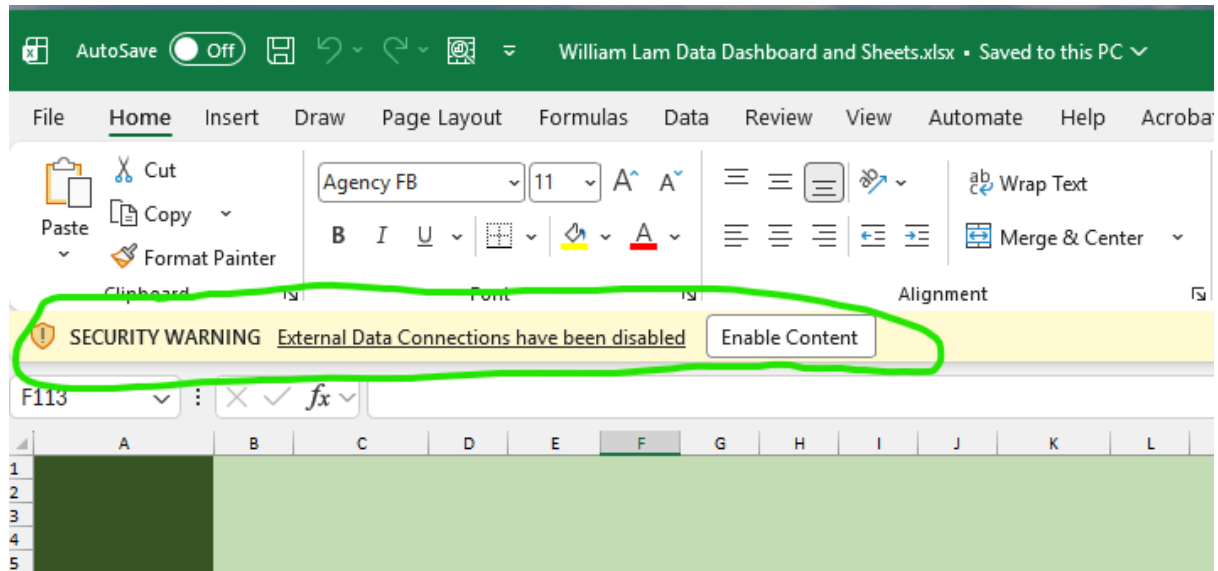
Follow steps:

<https://support.microsoft.com/en-au/topic/a-potentially-dangerous-macro-has-been-blocked-0952faa0-37e7-4316-b61d-5b5ed6024216>

Excel



Click Enable Editing.



Then Click Enable Content

All files should then be working properly.

Database

Schema

Overview

Entities:

- Countries
- Customers
- Products
- Regions
- Transactions

Table: Regions

PK: RegionID

Fields:

- RegionID
- RegionName

Table: Countries

PK: CountryID

FK: RegionID → Regions

Fields:

- CountryID
- CountryName
- RegionID

Table: Customers

PK: CustomerID

FK: CountryID → Countries

Fields:

- CustomerID
- CustomerNo
- CountryID

Table: Products

PK: ProductID

Fields:

- ProductID
- ProductNo
- ProductName

Bridge Table: Transactions

PK: TransactionID

FKs:

- CustomerID → Customers
- ProductID → Products

Fields:

- TransactionID
- InvoiceNo
- Date
- CustomerID
- ProductID
- Quantity
- Price
- UnitTax
- LineTotal

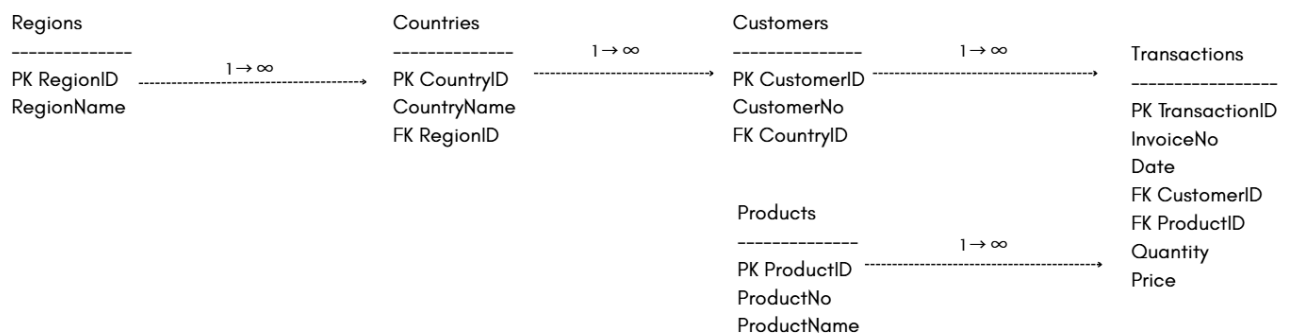
Entity Relationships

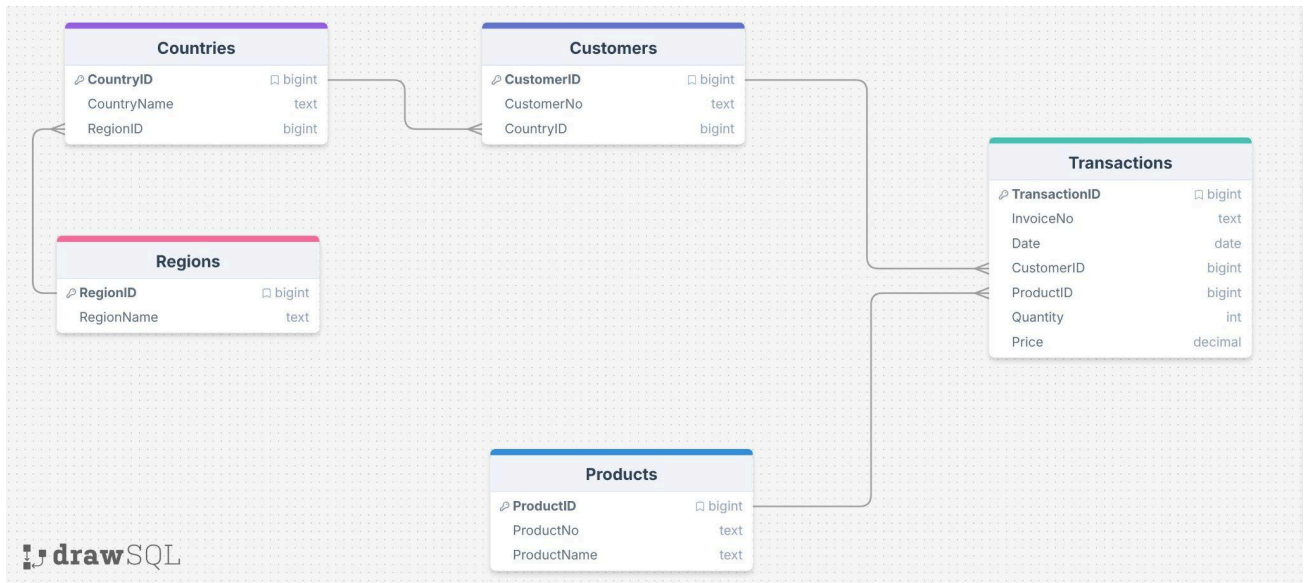
- Regions **1** → **many** Countries
 - *1 Region can have many Countries*
- Countries **1** → **many** Customers
 - *1 Country can have many Customers*
- Customers **1** → **many** Transactions
 - *1 Customer can have many Transactions*
- Products **1** → **many** Transactions
 - *1 Product can have many Transactions*

Transactions connect Customers and Products, hence it is:

- Customers **many** → **many** Products
 - *Many Customers can have many Products through Transactions*

ER Diagram





(Created using drawSQL - does not explicitly say 'one to many'. They are shown by the line dissecting into three parts for a 'many'.)

Data Dictionary

Regions Table

Field Name	Data Type	Data Formal	Field Size	Description	Example
RegionID	AutoNumber	NN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the region	1
RegionName	Short Text	XXXXX	20	Name of the shipping region; either domestic (shipping to UK) or international	Domestic

Countries Table

Field Name	Data Type	Data Formal	Field Size	Description	Example
CountryID	AutoNumber	NN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the country	12
CountryName	Short Text	XXXXXX	50	Name of the Country	Australia
RegionID	Number	NN	-2,147,483,648 to +2,147,483,647	Foreign Key from 'Regions' Table which identifies the region the country is in	1

Customers Table

Field Name	Data Type	Data Formal	Field Size	Description	Example
CustomerID	AutoNumber	NNNN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the customer	1000

CustomerNo	Short Text	XXXXX	15	Unique identifier given by the business to the customer so that it can be traced back to in the original dataset, as well as future datasets. It is a 'natural' key.	12004
CountryID	Number	NN	-2,147,483,648 to +2,147,483,647	Foreign Key from 'Countries' Table which identifies the country	36

Products Table

Field Name	Data Type	Data Formal	Field Size	Description	Example
ProductID	Autonumber	NNNN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the product	1000
ProductNo	Short Text	XXXXXX	20	Unique identifier given by the business to the transaction so that it can be traced back to in the original dataset, as well as future datasets. It is a 'natural' key.	22432A
ProductName	Short Text	XXXXX	255	Name of the Product	Cacti T-Light Candles

Transactions Table

Field Name	Data Type	Data Formal	Field Size	Description	Example
TransactionID	Autonumber	NNNNNN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the transaction	523215
InvoiceNo	Short Text	XXXXXXX	20	Unique identifier given by the business to the transaction so that it can be traced back to in the original dataset, as well as future datasets. It is a 'natural key'. InvoiceNos that begin with 'C' are cancellations.	C536365
Date	Date/Time	dd/mm/yyyy	8 bytes	Date the transaction was carried out	01/12/2018
CustomerID	Number	NNNN	-2,147,483,648 to +2,147,483,647	Foreign Key from the 'Customers' Table which identifies the customer	1234
ProductID	Number	NNNN	-2,147,483,648 to +2,147,483,647	Foreign Key from the 'Products' Table which identifies the product	3245

Quantity	Number	NNNNNN	-2,147,483,648 to +2,147,483,647	Quantity for products in the transaction	12345
Price	Currency	\$NNN.NN	8 bytes	Price of the product at the time of the transaction	\$23.15

Note: When importing the Access Database into Excel, I used a Power Query to create the 'UnitTax' and 'LineTotal' columns, as not to violate the 3NF principles in the database, as previously these columns in the database would have been dependencies on 'Quantity' and 'Price', which are not the primary key.